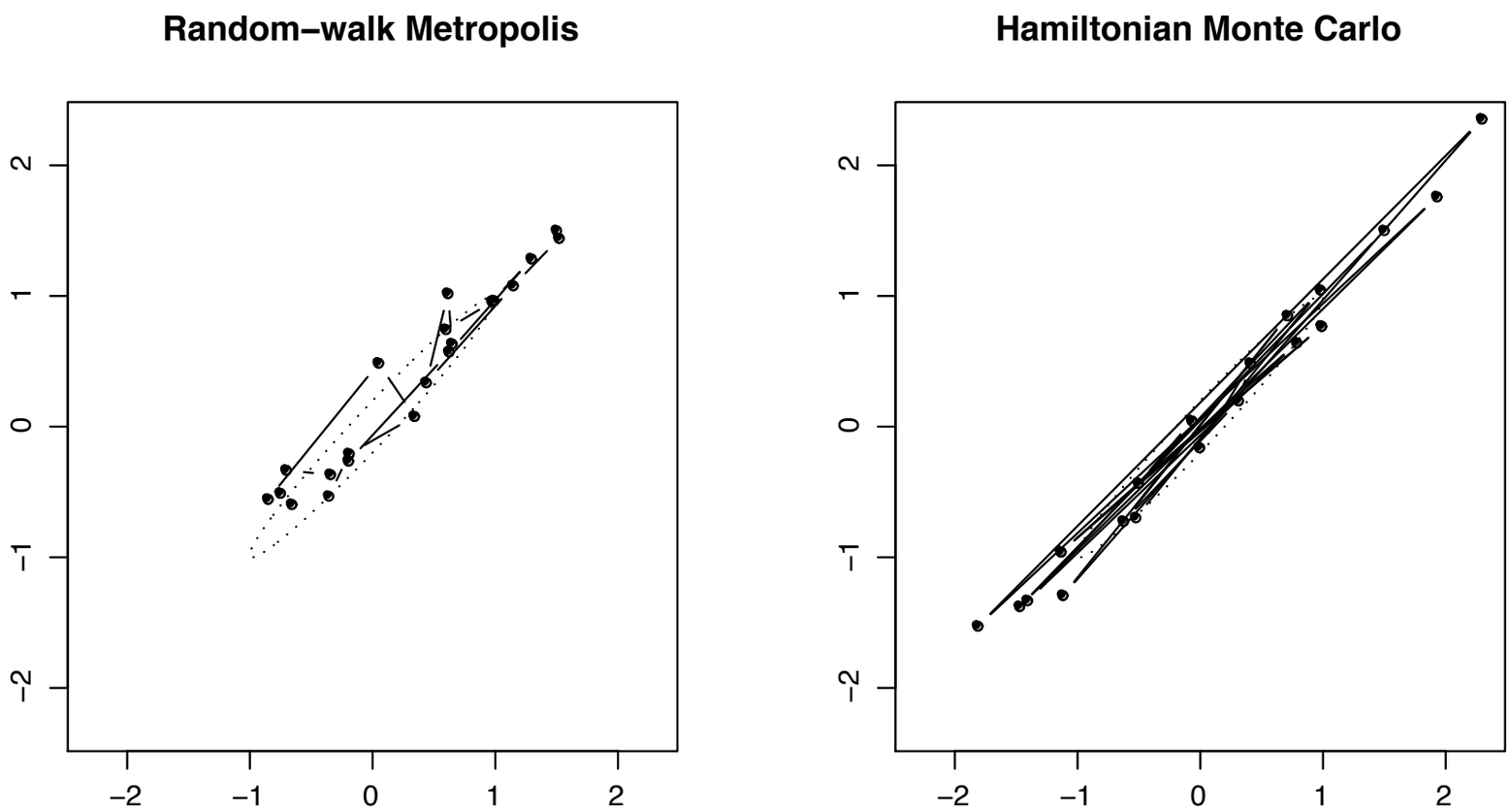
EXAMPLE

- Consider a 2D Gaussian distribution with covariance matrix $\Sigma = \begin{pmatrix} 1 & 0.98 \\ 0.98 & 1 \end{pmatrix}$
- We let $M = I$, and do $L = 25$ leapfrog steps per iteration using stepsize $\epsilon = 0.25$.
- Twenty iterations of the random walk Metropolis method (20 updates per iteration). HMC is making much larger moves and mixes faster than random walk Metropolis.



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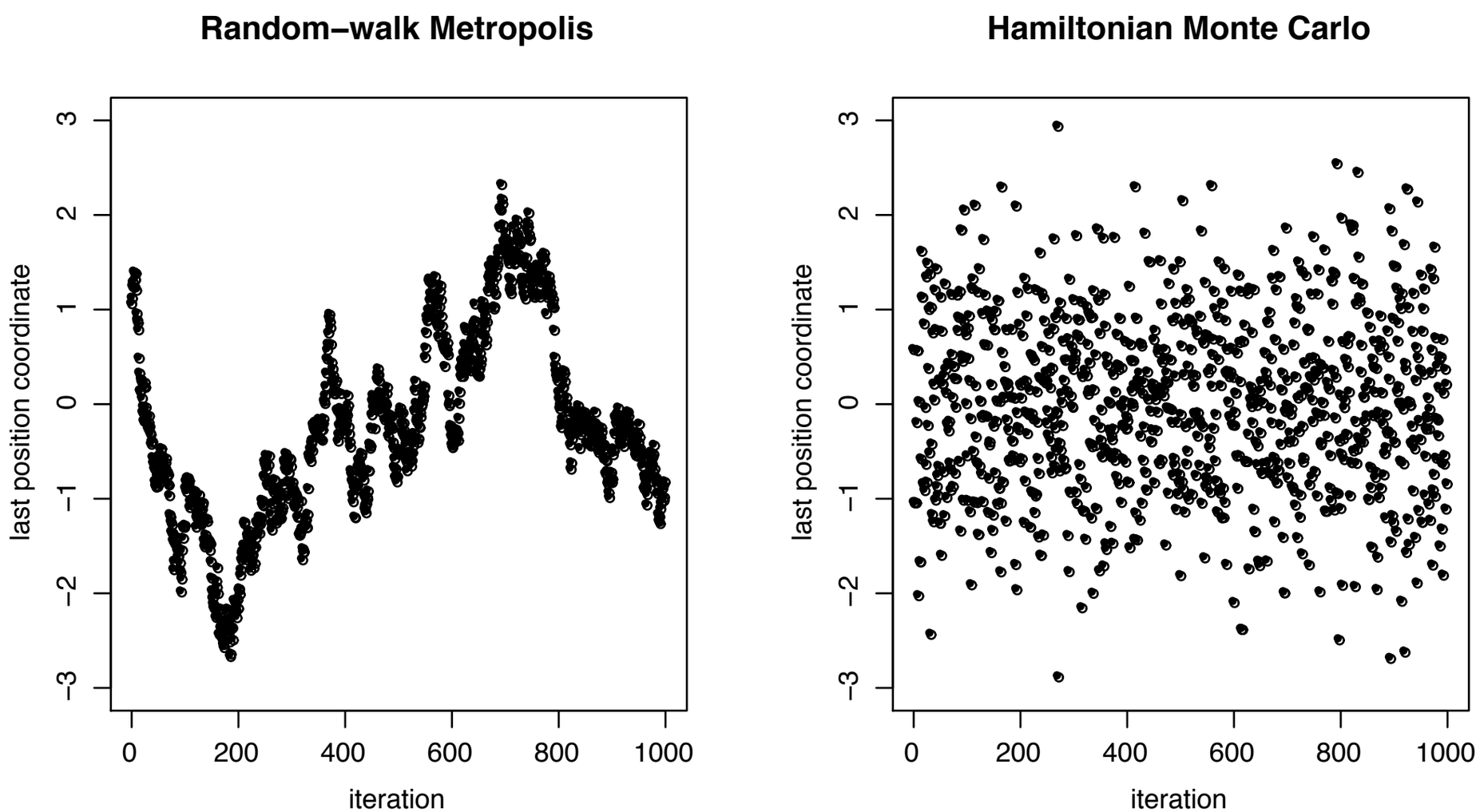
- 100 dimensional multivariate Gaussian distribution with independent components of standard deviations $0.01,\ldots,1.00$, with $\epsilon = 0.1$ and $L = 150$ leapfrog steps. The figure shows the last component.